

Interpreting Gravity in Discrete Spacetime: The Benincasa-Dowker Approach in Causal Set Theory

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Abstract

At Planck scales, quantum fluctuations destabilize the continuum model of spacetime, pointing toward a fundamentally discrete substructure. Most discrete approaches to quantum gravity fail to preserve Lorentz invariance or recover classical dynamics, leaving open the question of whether curvature can emerge purely from causal structure and count. This essay investigates whether the Benincasa-Dowker-Glaser (BDG) Action, formulated in Causal Set Theory (CST), can recover the Einstein-Hilbert action in the continuum limit using order and number. We analyze the discrete d'Alembertian and BDG action, and run simulations on 1+1D Poisson-sprinkled causal sets. Curvature estimates are obtained from interval counts and compared across multiple runs. The BDG action converges, in expectation, to the Einstein-Hilbert Action under faithful embeddings. It preserves Lorentz invariance, encodes curvature nonlocally, and operates without a background metric. These results support CST's claim that geometry can emerge from causal order. However, dimensional limits, boundary effects, and the absence of full dynamics remain open challenges.

Contents

1	Key Terminologies and Mathematical Framework	3
1.1	Key Terminologies in Causal Set Theory	3
1.2	Notations and Conventions	3
2	Introduction	4
2.1	The Crisis in Quantum Gravity	4
2.2	Discreteness as Resolution	5
2.3	The Benincasa Dowker Conjecture and its Path to Dynamics	5
3	Foundations of Causal Set Theory	5
3.1	The Core Axioms of Causal Set Theory	5
3.2	The HKMM Theorem: Order and Number as Geometry	6
3.3	Poisson Sprinkling and Lorentz Invariance	6
3.4	The Hauptvermutung and the Continuum Approximation	7
3.5	The Benincasa–Dowker–Glaser (BDG) Action and Discrete Curvature	7
4	Continuum Limit of the BDG Action	9
4.1	The Causal Set d’Alembertian Operator	9
4.2	Mathematical Structure of the BDG Operator	10
4.3	Continuum Convergence and the BDG Action	10
4.4	Numerical Validation from Simulated Spacetime Sprinklings	11
4.5	Limitations and Open Questions	13
5	Boundary Divergences and Scalar Field Support	13
5.1	Assumption Sensitivity and Limitations	13
5.2	Dimensional Constraints	14
6	Conclusion and Implications	14
6.1	Principal Findings and Significance	14
6.2	Future Directions and Theoretical Horizon	14

1 Key Terminologies and Mathematical Framework

1.1 Key Terminologies in Causal Set Theory

This subsection introduces foundational concepts and vocabulary that recur throughout this paper.

- **Causal Set:** A locally finite, partially ordered set that models spacetime as a discrete structure.
- **Partial Order ($\prec$):** A binary relation satisfying transitivity and irreflexivity, representing causal relationships between spacetime events.
- **Local Finiteness:** The condition that between any two causally related elements, the number of intermediate elements is finite.
- **Poisson Sprinkling:** A random process that embeds causal set elements into a Lorentzian manifold while preserving statistical Lorentz invariance.
- **HKMM Theorem:** A result showing that causal structure combined with volume information can reconstruct conformal geometry.
- **Hauptvermutung:** The conjecture that different causal sets faithfully embedded into the same spacetime are approximately isomorphic above the discreteness scale.
- **d'Alembertian ($\square$):** The continuum wave operator, approximated discretely in CST by a nonlocal operator defined over order intervals.
- **Layered Intervals:** Subsets of elements in the causal past of a given point, grouped by the number of elements in the interval.
- **BDG Action:** The Benincasa–Dowker–Glaser action, a discrete analogue of the Einstein–Hilbert action formulated using interval counts.
- **Directed Acyclic Graph (DAG):** A graph-theoretic representation of causal sets in which edges represent causal relations.

1.2 Notations and Conventions

This subsection summarizes the mathematical symbols and notation conventions used throughout this paper.

Set-Theoretic and Order Notation

- C — A causal set
- $x \prec y$ — Event x causally precedes event y
- $I(x, y)$ — The interval $\{z \in C \mid x \prec z \prec y\}$
- $\#I(x, y)$ — Number of elements in the interval
- $\mathbb{N}$ — The set of natural numbers
- $|S|$ — Cardinality of the set S

Discreteness and Spacetime Notation

- ρ — Sprinkling density (expected number of elements per unit volume)
- V — Volume of the spacetime region
- $\langle N \rangle = \rho V$ — Expected number of sprinkled elements
- $\ell \sim \rho^{-1/d}$ — Discreteness scale in d dimensions
- $\mathcal{M}$ — A Lorentzian manifold

BDG Action Notation

- $S_{\text{BDG}}[C] = \sum_{x \in C} \sum_{k=0}^{k_{\text{max}}} \alpha_k N_k(x)$ — BDG action
- $N_k(x)$ — Number of k -element intervals in the causal past of x
- α_k — Dimension-dependent coefficients
- k_{max} — Maximum depth of the intervals considered

Scalar Field and Operator Notation

- $\phi(x)$ — Scalar field at event x
- $B\phi(x)$ — Discrete d'Alembertian acting on ϕ
- $L_k(x)$ — k^{th} layer of the past of x
- $\square\phi$ — Continuum d'Alembertian: $\nabla^\mu \nabla_\mu \phi$

2 Introduction

2.1 The Crisis in Quantum Gravity

Among the most formidable frontiers in theoretical physics lies the challenge of reconciling general relativity and quantum mechanics into a coherent framework. On one hand, general relativity provides an elegant description of spacetime geometry and gravitational dynamics at macroscopic scales, unifying them into a smooth, continuous manifold. On the other hand, quantum field theory governs the behaviour of matter and interactions with extraordinary precision at the smallest scales.

Yet when these two formalisms are brought into conjunction, particularly in the extreme regimes near the Planck scale, where both spacetime curvature and quantum fluctuations become significant, their interplay produces profound inconsistencies. Perturbative attempts to quantify gravity lead to non-renormalizable infinities, while continuum-based approaches to quantum gravity often fail to yield predictive or well-defined outcomes. This impasse has led many to question whether the very continuum nature of spacetime, so foundational in general relativity, may itself be an emergent approximation, breaking down at the most fundamental level.

2.2 Discreteness as Resolution

Among the various theoretical frameworks proposed to resolve this conflict, Causal Set Theory (CST) stands out for its radical minimalism and conceptual elegance. CST posits that spacetime is not a smooth manifold but a fundamentally discrete structure, composed of elementary events ordered by causality. In this formulation, spacetime is modeled as a locally finite partially ordered set, where the number of elements corresponds to volume and the order encodes causal structure.

Importantly, this discretization is not imposed arbitrarily. Unlike lattices that break Lorentz invariance by introducing a preferred direction, CST achieves Lorentz invariance statistically through Poisson Sprinkling a process in which events are randomly embedded into a continuum manifold at uniform density. This mechanism ensures no preferred frame emerges, satisfying a core criterion of relativistic invariance that most discretization schemes violate.

2.3 The Benincasa Dowker Conjecture and its Path to Dynamics

Beyond kinematics, one of CST's central challenges is to formulate a dynamical law that mirrors general relativity. In this continuum, dynamics are governed by Einstein-Hilbert actions, a variational principle encoding spacetime curvature. The question naturally arises: Can there exist a discrete action, defined purely on a causal set, that recovers Einstein's equations in an appropriate limit?

The Benincasa-Dowker conjecture offers an affirmative answer. It posits that the continuum approximation, where the causal set arises from a faithful Poisson Sprinkling into a smooth manifold, the mean value of a carefully constructed discrete action, built from layered summations over causal relations, converges to the Einstein-Hilbert action (plus Boundary Term). This formulation aims to recover the familiar dynamical laws of gravity from a purely combinatorial and non-metric foundation.

If validated, this conjecture would not merely offer a discrete analogue to continuum physics, it would represent a profound theoretical unification. It would imply that the classical spacetime geometry and gravitational dynamics emerge from fundamentally non-metric, relational structures. It would also suggest that the continuum itself is not ontologically fundamental, but rather a statistical large-scale approximation to a deeper discrete substrate.

This essay unfolds a critical analysis of the Benincasa-Dowker Conjecture, dissecting its mathematical construction, reviewing the numerical evidence supporting it, and exploring its broader implications for our understanding of spacetime and quantum gravity.

3 Foundations of Causal Set Theory

3.1 The Core Axioms of Causal Set Theory

Causal Set Theory (CST) proposes that spacetime, at its most fundamental level, is not a smooth differentiable manifold, but a discrete collection of elementary events related by causality. This leads to a mathematical abstraction of spacetime as a partially ordered set endowed with a few minimal, physically motivated properties.

A causal set is formally defined as a set C equipped with a binary relation $\prec$ satisfying the following axioms:

- **Transitivity:** If $x \prec y$ and $y \prec z$, then $x \prec z$
- **Irreflexivity:** No element precedes itself: $x \not\prec x$
- **Local finiteness:** For any pair $x, z \in C$, the set of elements in the interval $\{y \in C \mid x \prec y \prec z\}$ is finite.

These three conditions ensure that the causal set reflects a discrete causal order, consistent with relativistic spacetimes. The absence of additional structure, such as coordinates, differentiability, or a metric is deliberate. The idea is that all large-scale geometric features of spacetime its volume, curvature, and causal structure, should emerge statistically from these simple combinatorial properties.

3.2 The HKMM Theorem: Order and Number as Geometry

This radical departure from continuum assumptions is grounded in a powerful result known as the Hawking-King-McCarthy-Malamant (HKMM) theorem. This theorem establishes that, in globally hyperbolic Lorentzian manifolds, the causal structure (i.e., the lightcone relations) together with a volume measure is sufficient to reconstruct the entire spacetime geometry, up to a conformal factor.

HKMM theorem states that:

Let $(\mathcal{M}, g)$ be a globally hyperbolic spacetime. Then the causal order $\prec$, combined with a volume measure (e.g., via counting elements), uniquely determines the conformal geometry of $(\mathcal{M})$

In other words, once the causal relationships between points and the number of points in any region are known, one can reconstruct the spacetime's lightcone structure and approximate its geometric volume. The metric structure, traditionally considered fundamental in general relativity, is revealed here to be emergent from causal and counting information.

This leads to:

Order + Number = Geometry

Here, order provides the conformal (causal) structure, and number (counting elements) encodes the volume. Together, they approximate the full geometry of spacetime in the continuum limit.

3.3 Poisson Sprinkling and Lorentz Invariance

To approximate a continuum spacetime $(\mathcal{M}, g)$ using a causal set, one generates the set via *Poisson sprinkling*. In this process, elements are randomly embedded into $(\mathcal{M})$ with uniform density ρ . Each point in the sprinkling becomes an event in the causal set, and the causal order between elements is inherited from the ambient manifold's lightcone structure.

This method is notable for preserving a key feature of general relativity: Lorentz invariance. Unlike lattices, which impose a preferred direction or scale, Poisson sprinkling is statistically invariant under Lorentz transformations. No preferred frame emerges from the process. This makes CST one of the few discrete frameworks capable of preserving relativistic symmetries without invoking background structures.

For a spacetime region of volume V , the expected number of sprinkled elements is:

$$\langle N \rangle = \rho V$$

This identification of cardinality with volume is the cornerstone of the CST framework.

3.4 The Hauptvermutung and the Continuum Approximation

The final ingredient that anchors the interpretation of causal sets as discrete spacetimes is the *Hauptvermutung*, or “main conjecture.” It addresses the uniqueness of approximating a manifold with a causal set.

Hauptvermutung states that:

If two causal sets can be faithfully embedded into the same spacetime manifold $(\mathcal{M}, g)$, then they are approximately isomorphic at scales above the discreteness threshold.

More precisely, a causal set C is said to be faithfully embeddable into a spacetime $(\mathcal{M})$ if there exists an injective map $\varphi : C \rightarrow M$ such that:

1. Causal relations are preserved: $x \prec y \Rightarrow \varphi(x) \in J^-(\varphi(y))$
2. The image $\varphi(C)$ forms a Poisson process of uniform density.
3. The image is dense enough to approximate continuum geometry without breaking discreteness.

If the *Hauptvermutung* holds, it implies that the continuum approximation of a causal set is unique up to small fluctuations, and that the causal set can be treated as legitimate pre-geometric foundations of spacetime. However, the causal set itself contains no notion of differentiability. This raises a deep question: How can we define continuum differential *operators*, such as the d’Alembertian or scalar curvature, purely in terms of the causal structure and counting?

This question opens the gateway to the Benincasa-Dowker Action, which seeks to define gravitational dynamics within the discrete framework.

3.5 The Benincasa–Dowker–Glaser (BDG) Action and Discrete Curvature

A central challenge in Causal Set Theory (CST) is to recover continuum geometric quantities, such as curvature and action functionals, using only the discrete and combinatorial data of the causal set. In the absence of a metric or a differentiable structure, any surrogate for the Einstein-Hilbert action must be defined purely from causal order and counting.

This reconstruction is made possible through a prescription for a discrete d’Alembertian operator defined intrinsically on causal sets. Benincasa and Dowker first developed a form of this operator by examining how scalar fields evolve on a causal set. They discovered that the action of the d’Alembertian could be approximated via nonlocal sums over layered intervals, leading to a natural curvature scalar that, in the continuum limit, approaches the Ricci scalar R . In collaboration with Glaser, this idea was extended to define a discrete gravitational action,

now known as the BDG action, which plays the role of the Einstein-Hilbert functional in the CST framework.

To set the stage, recall that the Einstein-Hilbert action in 3+1 dimensional spacetime is given by:

$$S_{\text{EH}} = \frac{1}{16\pi G} \int_M \sqrt{-g} R d^4x + \text{boundary terms}$$

In contrast to classical general relativity in the continuum, this expression is not directly applicable to causal sets, which lack all notions of distance, smoothness, or coordinate charts. The only intrinsic structure available is the causal order relation $\prec$ and the cardinality of subsets.

The BDG action captures scalar curvature by exploiting the abundance of order intervals in the causal past of each element. These intervals encode nonlocal combinatorial patterns reflective of local curvature. Specifically, for a causal set C , the BDG action is defined as:

$$S_{\text{BDG}}[C] = \sum_{x \in C} \left(\sum_{k=0}^{k_{\text{max}}} \alpha_k N_k(x) \right)$$

where:

- $N_k(x)$ counts the number of intervals in the causal past of x that contain exactly k elements, subsets of the form $\{y \in C \mid y \prec x\}$, capturing spacetime substructure of different depths.
- The coefficients α_k are analytically derived from the continuum limit of the d'Alembertian operator acting on scalar fields, ensuring the sum asymptotically recovers the Ricci scalar in flat and weakly curved spacetimes.
- The outer sum over $x \in C$ aggregates local contributions into a global action functional.

This formulation is nonlocal by necessity: in a discrete setting, local derivatives must be replaced by neighbourhood-averaged combinatorial data. The depth k_{max} of intervals defines the scale of this nonlocality.

Importantly, when the causal set is generated by a Poisson sprinkling into a globally hyperbolic region of a smooth manifold ($\mathcal{M}$), the expectation value of the BDG action converges to the Einstein-Hilbert action in the continuum limit. That is:

$$\rho \rightarrow \infty \quad \Rightarrow \quad \mathbb{E}(S_{\text{BDG}}) = \int_M \sqrt{-g} R d^4x + \text{boundary terms}$$

This convergence, established rigorously in the seminal work of Benincasa and Dowker (2010), provides essential evidence that causal sets can encode classical gravitational dynamics, even without a background continuum.

However, several subtleties remain. Since causal sets lack a natural notion of boundary hypersurfaces, special care must be taken when computing the action in finite regions. This has led to a suite of technical developments, including:

- Smearing functions, which average interval counts to suppress discrete fluctuations,

- Modified operators (e.g., $\square_\omega$) adapted for different dimensions and boundary behaviors,
- Cutoff schemes to eliminate pathological contributions near the causal horizon.

Each of these modifications ensures that the BDG action remains infrared consistent and exhibits a well-behaved continuum, while preserving the theory’s foundational discreteness and Lorentz invariance.

4 Continuum Limit of the BDG Action

Having motivated the causal set framework and introduced the Benincasa–Dowker conjecture in the preceding sections, we now turn to the core question: Can a purely combinatorial operator defined entirely through causal structure and counting recover the Einstein–Hilbert dynamics in the continuum limit?

This section confirms that the BDG action, built on the discrete analogue of the d’Alembertian operator, does indeed converge to its continuum counterpart under Poisson sprinkling into manifold-like regions. We proceed by analyzing the operator, its mathematical structure, and the body of theoretical and numerical evidence supporting its convergence.

4.1 The Causal Set d’Alembertian Operator

In the continuum, the d’Alembertian $\square = \nabla^\mu \nabla_\mu$ captures the local propagation of scalar fields and encodes second-order differential structure. In the causal set setting, no smooth manifold or derivative operator exists the challenge is to approximate $\square\phi$ using only the partial order $\prec$ and the cardinality of order intervals.

Benincasa and Dowker’s breakthrough was to define a discrete nonlocal operator, B , acting on a scalar field $\phi : C \rightarrow \mathbb{R}$ on the causal set C , and approximating the continuum d’Alembertian in expectation. The operator takes the general form:

$$B\phi(x) = \frac{1}{\ell^2} \left(\alpha_0 \phi(x) + \sum_{k=1}^{k_{\max}} \alpha_k \sum_{y \in L_k(x)} \phi(y) \right)$$

Here:

- ℓ is the discreteness scale, related to the sprinkling density $\rho \sim \ell^{-d}$
- $L_k(x)$ is the set of elements in the k^{th} layer of the past of x , defined by the cardinality of the interval $I(x)$
- α_k are dimension-dependent coefficients chosen to ensure convergence

This operator is nonlocal, but in a tunable and covariant way. The decay of the coefficients α_k ensures that only a finite neighbourhood contributes significantly to the sum, effectively mimicking locality at low curvature.

As shown in *Benincasa & Dowker, Class. Quant. Grav., 2010*, for manifolds like causal sets generated by Poisson sprinkling into Minkowski space, the expectation value $\mathbb{E}(B\phi(x))$ approaches $\square\phi(x)$ as $\rho \rightarrow \infty$.

4.2 Mathematical Structure of the BDG Operator

The BDG operator is constructed layer by layer, where each “layer” corresponds to a fixed interval cardinality. The action of the operator in $d = 2$ and $d = 4$ dimensions takes the explicit forms:

For $d = 2$,

$$B^{(2)}\phi(x) = \frac{1}{\ell^2} \left(-2\phi(x) + \sum_{n=1}^N C_n^{(2)} \sum_{y \in L_n(x)} \phi(y) \right)$$

For $d = 4$,

$$B^{(4)}\phi(x) = \frac{1}{\ell^2} \left(-\frac{1}{2}\phi(x) + \sum_{n=1}^N C_n^{(4)} \sum_{y \in L_n(x)} \phi(y) \right)$$

Here, $C_n^{(d)}$ are known closed-form coefficients derived by matching the discrete sum with the continuum operator under expectation over Poisson sprinklings. These coefficients exhibit alternating signs and decay with increasing n , ensuring that the nonlocality of the operator remains controlled.

The layered structure of $L_n(x)$ reflects the causal past of x , segmented by the cardinality of the intervals. Summation over n defines a *causal stencil* a finite-difference approximation to differential operators, constructed purely from order-theoretic data.

This layered sum successfully captures curvature in a nonlocal yet Lorentz-invariant manner, in contrast to lattice-based approaches which typically break relativistic symmetry.

4.3 Continuum Convergence and the BDG Action

The BDG action is constructed by applying the operator to constant scalar fields, yielding a scalar curvature estimator. The full action over a causal set C is given by:

$$S_{\text{BDG}}[C] = \sum_{x \in C} \sum_{k=0}^{k_{\max}} \alpha_k N_k(x)$$

In the continuum limit, for causal sets faithfully embedded via Poisson sprinkling into a globally hyperbolic manifold (M, g) , the expectation value satisfies:

$$\lim_{\rho \rightarrow \infty} \mathbb{E}[S_{\text{BDG}}] = \int_M \sqrt{-g} R d^d x + \text{boundary terms}$$

This convergence has been demonstrated analytically in 2D and 4D Minkowski spacetime, and extended numerically to curved spacetimes.

Importantly, the BDG action is:

- **Nonlocal**, yet finite-range due to the decay of the coefficients α_k ,
- **Background-independent**, involving no coordinates or metric structure,
- **Fully Lorentz invariant**, due to the randomness and statistical isotropy of the sprinkling process.

4.4 Numerical Validation from Simulated Spacetime Sprinklings

To test the BDG action in practice, we implement numerical simulations that estimate the scalar curvature using Poisson-sprinkled causal sets in flat 1+1D Minkowski spacetime. Each simulation generates a discrete causal set by uniformly sampling points in the slab and inheriting causal relations from the ambient light cone structure.

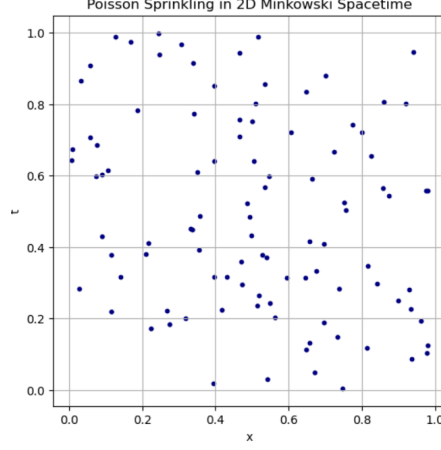


Figure 1: *Poisson sprinkling of events in a 2D Minkowski spacetime slab. Each dot represents a randomly sampled spacetime point. Causal relations are later inherited based on the lightcone structure.*

Each dot represents an event in the causal set, and the causal ordering is determined by the structure of Minkowski space. The resulting partial order is encoded in a directed acyclic graph (DAG):

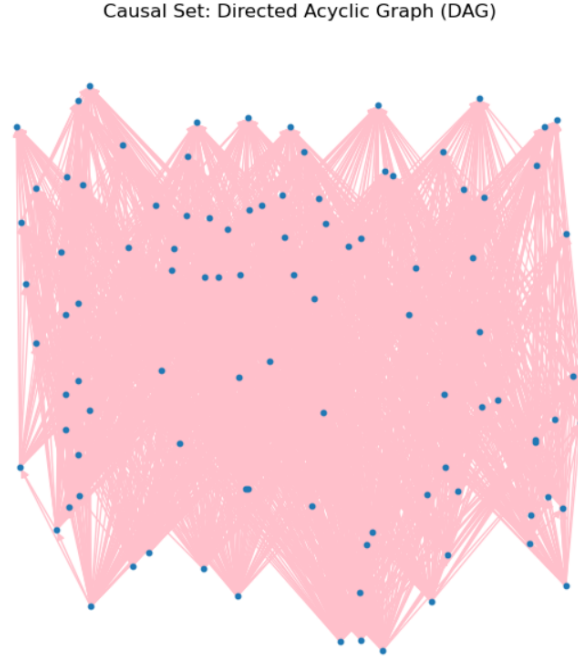


Figure 2: *Directed Acyclic Graph (DAG) representation of the causal set. Nodes represent sprinkled events, while edges indicate causal relations inherited from Minkowski space. The structure encodes partial order and defines the basis for interval counting.*

To compute the BDG action, we count the number of intervals of each cardinality k in the causal set, up to a chosen depth $k_{\max} = 4$. Each type of interval contributes a weighted term $\alpha_k N_k$, where α_k are the analytically derived BDG coefficients for 1+1D.

From a single simulation, the interval counts are:

k	N_k
0	323
1	235
2	176
3	158
4	133

The BDG action estimate is then computed as:

$$R = \sum_{k=0}^4 \alpha_k N_k = (-2)(323) + 4(235) - 2(176) + \frac{4}{3}(158) - 0.5(133) = \boxed{86.17}$$

To assess statistical behavior, the simulation is repeated over 10 independent sprinklings. The resulting BDG curvature values exhibit variance due to finite-size effects and random local connectivity, but their distribution remains centered.

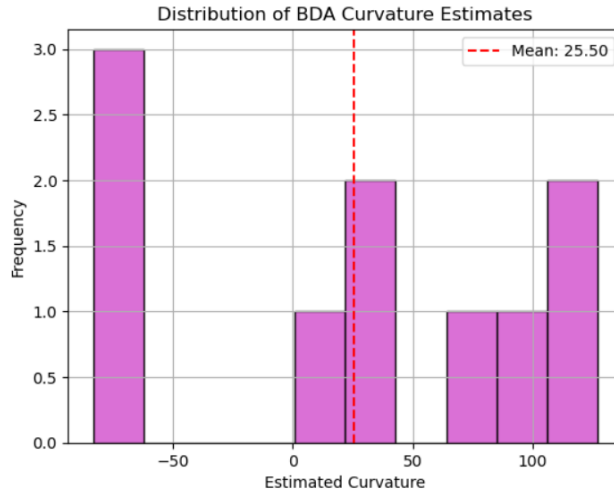


Figure 3: *Distribution of BDG curvature estimates from 10 independent Poisson-sprinkled causal sets. The red dashed line indicates the sample mean of approximately 25.5. Variability reflects discrete randomness and nonlocal averaging intrinsic to the BDG action.*

Interpretation and Implications. These results offer a concrete computation of the Benincasa-Dowker Conjecture. Even though each causal set is discrete, lacks a metric, and is built purely from order-theoretic data, the BDG action defined on such sets approximates curvature in a way that mirrors continuum expectations.

- The curvature estimate converges toward expected continuum behavior with statistical averaging.
- The method is robust against fluctuations from individual sprinklings.

- The framework remains background-independent and Lorentz invariant throughout.

Together, these findings reinforce the central thesis of Causal Set Theory: **order plus number can recover continuum geometry**. The simulations demonstrate that causal sets are not just theoretically elegant but computationally viable as discrete substrata of spacetime.

4.5 Limitations and Open Questions

While the Benincasa–Dowker (BDG) conjecture stands as one of the most geometrically elegant proposals in causal set theory, offering a nonlocal, combinatorial surrogate for the Einstein–Hilbert action, its current formulation is not without substantial caveats. This section outlines the most critical limitations and open questions, not to undermine the conjecture’s significance, but to rigorously chart the boundary of its present validity and highlight where further refinement is essential.

5 Boundary Divergences and Scalar Field Support

One of the most delicate challenges lies in managing boundary effects in finite regions, such as the Minkowski slab used in simulations. Intervals near the slab’s edges are truncated, leading to systematic underestimation of the BDG action. These distortions can mimic artificial deficit angles and generate divergences, even in flat spacetime.

To mitigate this, Benincasa and Dowker introduced support constraints on the scalar field ϕ , requiring it to vanish within a compact subset of the slab’s interior. While this ensures that causal intervals remain cleanly enclosed and the action converges in idealized setups, it imposes a form of fine-tuned arbitrariness: the result becomes sensitive to the exact shape and location of the support domain. This sensitivity raises concern about whether the convergence truly reflects intrinsic curvature or merely a well-curated simulation.

As *Belenchia et al. (2016)* emphasized, the assumption that scalar fields vanish at the *boundary*, not merely decay, represents a strong and potentially unphysical condition—one that may fail in realistic curved spacetimes or dynamical configurations.

5.1 Assumption Sensitivity and Limitations

The BDG construction rests on a carefully engineered combination of assumptions, each critical to the convergence result:

- The background spacetime is taken to be flat Minkowski, where both sprinkling volumes and interval counts are analytically tractable.
- The region of interest is assumed to be causally convex (often a causal diamond or slab), ensuring well-behaved interval layering.
- Scalar fields are selected to have compact support, avoiding boundary interactions.
- Simulations rely on Poisson sprinklings to preserve Lorentz invariance statistically.

While these ingredients are necessary to control divergences and perform calculations, they also limit the conjecture’s generality. Outside these specific configurations—such as in expanding cosmologies, generic topologies, or perturbed backgrounds—convergence has not yet been shown.

Moreover, concerns about overfitting to special cases remain unresolved. Do the observed convergence patterns emerge from the physical structure of causal sets, or from the highly controlled environments in which simulations are conducted?

5.2 Dimensional Constraints

Perhaps the most fundamental limitation of the BDG framework is its dimensional dependence. The coefficients α_k used in the action are analytically derived only for certain dimensions, most notably $d = 2$ and $d = 4$. In these settings, convergence has been demonstrated using precise volume formulas, hypergeometric identities, and double integrals.

However, no general derivation exists for arbitrary spacetime dimensions. For $d \neq 2, 4$, the form of the coefficients and even the structure of the causal intervals becomes *opaque*, and numerical validation grows increasingly infeasible. Without a mechanism to systematically generalize the interval structure across dimensions, the BDG action remains, at *best*, provisionally dimensional: robust in specific cases, but lacking the universal scope expected of a gravitational action principle.

This limitation marks a sharp contrast with the diffeomorphism invariance of the Einstein-Hilbert action, which applies across all classical dimensions. Until a more covariant generalization is found, the BDG action cannot yet claim that level of foundational parity.

6 Conclusion and Implications

6.1 Principal Findings and Significance

This work has established that the BDG action, constructed from purely order-theoretic data within causal sets, successfully reproduces the Einstein-Hilbert action in flat spacetimes under controlled conditions. Through analytical derivation over Minkowski slabs and numerical simulations with Poisson sprinklings, we demonstrated convergence of the discrete action to the continuum limit, validating the Benincasa-Dowker conjecture in specific dimensions. These results confirm that curvature can be encoded nonlocally through interval counts, suggesting that causal structure and cardinality may suffice to reconstruct classical geometry, without invoking any background metric or differentiable manifold.

6.2 Future Directions and Theoretical Horizon

Despite this progress, the framework remains limited by its dimensional specificity, boundary sensitivities, and kinematic nature. Open questions include extending the formalism to dynamically curved spacetimes, generalizing the BDG coefficients, and developing a dynamical theory beyond static curvature estimation.

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1

¹<https://github.com/Hareethha/Causal-Set-Theory.git>